Symmetry and skew

1

a Symmetrical

b Negative skew/Left skew

c Uniform distribution

d Clusters and gaps

e Negative skew/Left skew

f Positive skew/Right skew

g Negative skew/Left skew

h Positive skew/Right skew

i Symmetrical

j Symmetrical

k Negative skew/Left skew

l Positive skew/Right skew

2

a No **b** No **c** Yes **d** No

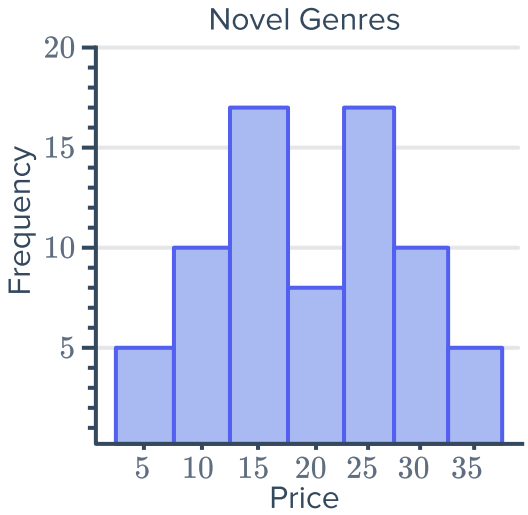
e Yes **f** Yes **g** No **h** Yes

i Yes **j** No

3 Multimodal, there are 3 peaks.

4

a



b Symmetrical and bimodal.

Clusters and outliers

5

a 2 **b** No **c** 26 and 36

6

a 92

b Yes, 30s – 40s

c 48

d Negative skew/Left skew

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7

a No

b Yes, $45 - 49$

c 47

d Negative skew/Left skew

8

a

Score	Frequency
$40 - 49$	3
$50 - 59$	7
$60 - 69$	3
$70 - 79$	5
$80 - 89$	8
$90 - 99$	4
$100 - 109$	8
$110 - 119$	4

b Multimodal with 3 peaks.

c $50 - 59$, $80 - 89$ and $100 - 109$

9

a Bimodal

b $1\% - 2\%$ and $3\% - 4\%$

10

a No

b No

c 15 and 21

d Symmetrical and bimodal.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

11

a

Score	Frequency
0.6	1
0.7	3
0.8	4
0.9	10
1	8
1.1	6
1.2	7
1.3	10
1.4	7
1.5	8
1.6	10
1.7	7
1.8	3

b Multimodal, with 3 peaks.

c $0.9, 1.3$ and 1.6

12

a No

b Yes, $1 - 4$

c 3

d Positive skew/Right skew

13

a No

b Yes, $9 - 10$

c 9

d Negative skew/Left skew

14 12.5

15

a Grouped data

b 151.5 cm

c Negative skew/Left skew

16

a Negative skew/Left skew

b

i 6 **ii** 8

c 2

d No

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

17

a Positive skew/Right skew

b

i 2 **ii** 4

c 2

d Yes, 9

18

a 18 years old

b 8 years

c 36 years

d 21.74 years old

e Positively skewed

19

a 30

b 42.5

c 26.5

d Positively skewed

e 76

f They do none to moderate amounts of exercise.

20

a **A** On each roll, each number is equally likely to appear.

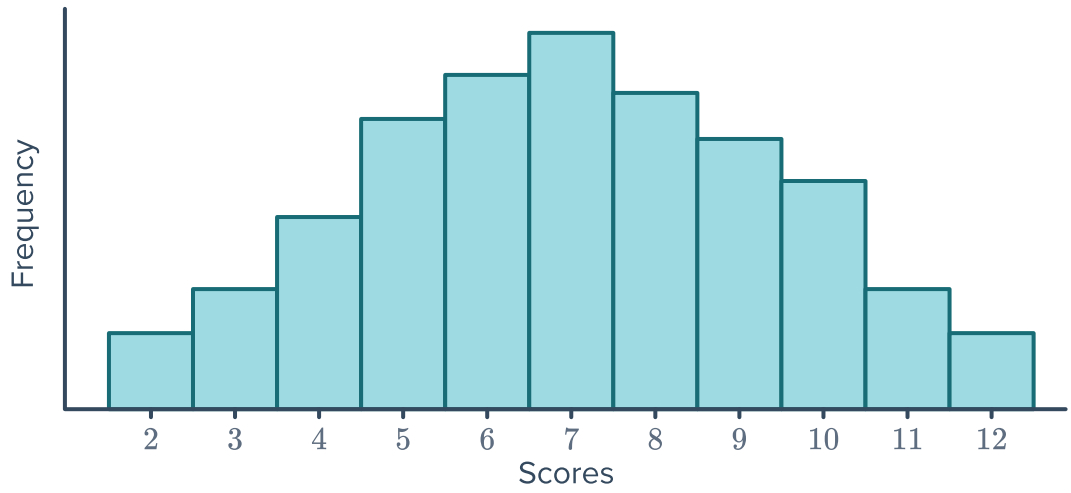
b Uniformly distributed

21

a

i 3 **ii** 6 **iii** 3

b



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

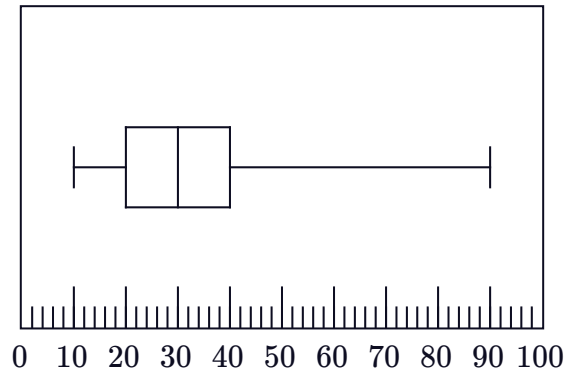
Connect histograms and box plots

22

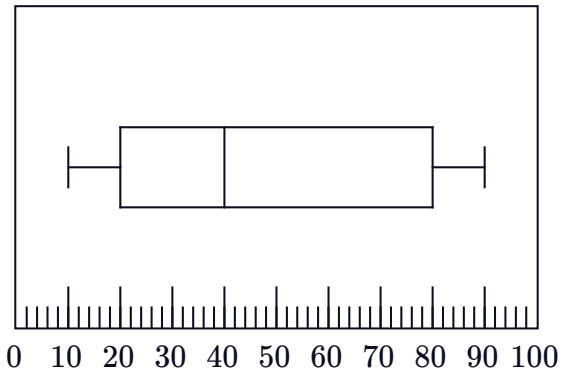
- Box Plot 1/Histogram E
- Box Plot 2/Histogram A
- Box Plot 3/Histogram B
- Box Plot 4/Histogram D
- Box Plot 5/Histogram F
- Box Plot 6/Histogram C

23

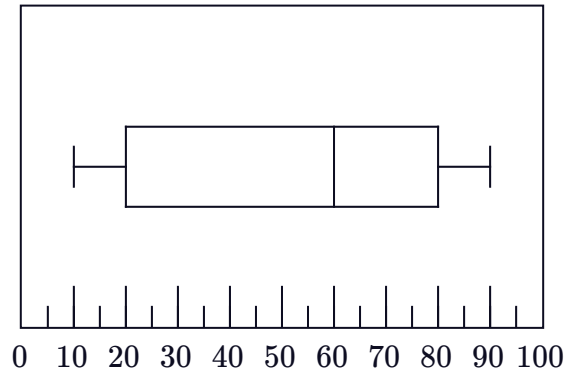
a



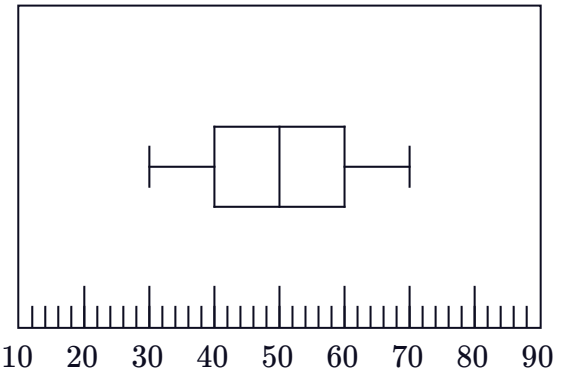
b



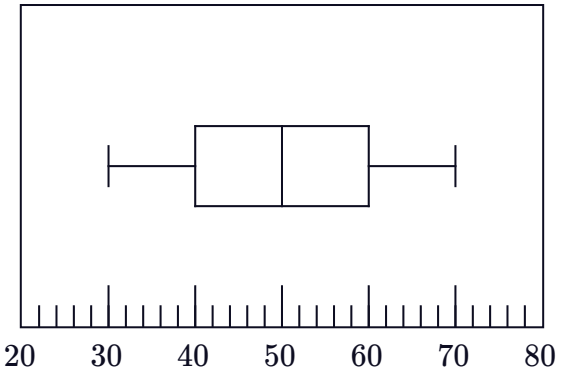
c



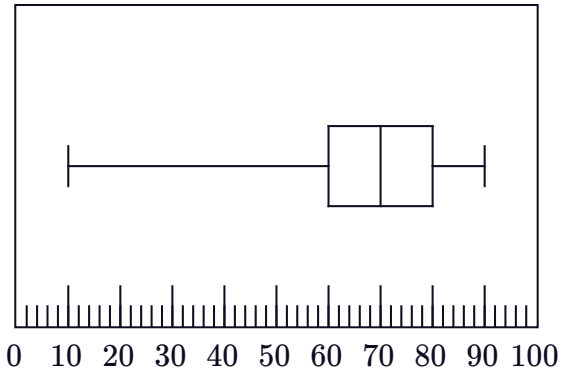
d



e



f



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

24

a Yes **b** No **c** No **d** Yes

e Yes

25

a The box plot has a long tail to the right which indicates positive skew, while the histogram does not appear to be skewed. The median for the histogram is roughly in the middle, while the median of the box plot is located further to the left.

b The box plot indicates that the data is widely spread, while the histogram indicates that most of the data occurs near the median.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647